WORKSHEET

Chapter 10: Normal distributions

1 Find the 85% confidence interval for a standard normal distribution.

2 Find the 98% confidence interval for a standard normal distribution.

3 Find the 70% confidence interval for a standard normal distribution.

4 Find the 20% confidence interval for a standard normal distribution.

5 Find the 80% confidence interval for the normal variable X with mean 100 and standard deviation 25.

6 Find the 96% confidence interval for the normal variable X with mean 58 and standard deviation 11.

7 Find the 99% confidence interval for the normal variable X with mean 75 and standard deviation 12.

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8 What is the confidence level for the interval 50–100 for a normal variable with mean 75 and standard deviation 9?

9 What is the confidence level for the interval 110–180 for a normal variable with mean 145 and standard deviation 22?

10 What is the confidence level for the interval 134–196 for a normal variable with mean 165 and standard deviation 32?

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Answers

1 About −1.44 to 1.44.

2 About −2.33 to 2.33.

3 About −1.04 to 1.04.

4 About −0.25 to 0.25.

5 About 32.04 to 167.96.

6 About 22.59 to 93.41.

7 About 30.91 to 119.09.

8 About 99.45%

9 About 88.84%

10 About 57.90%

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